



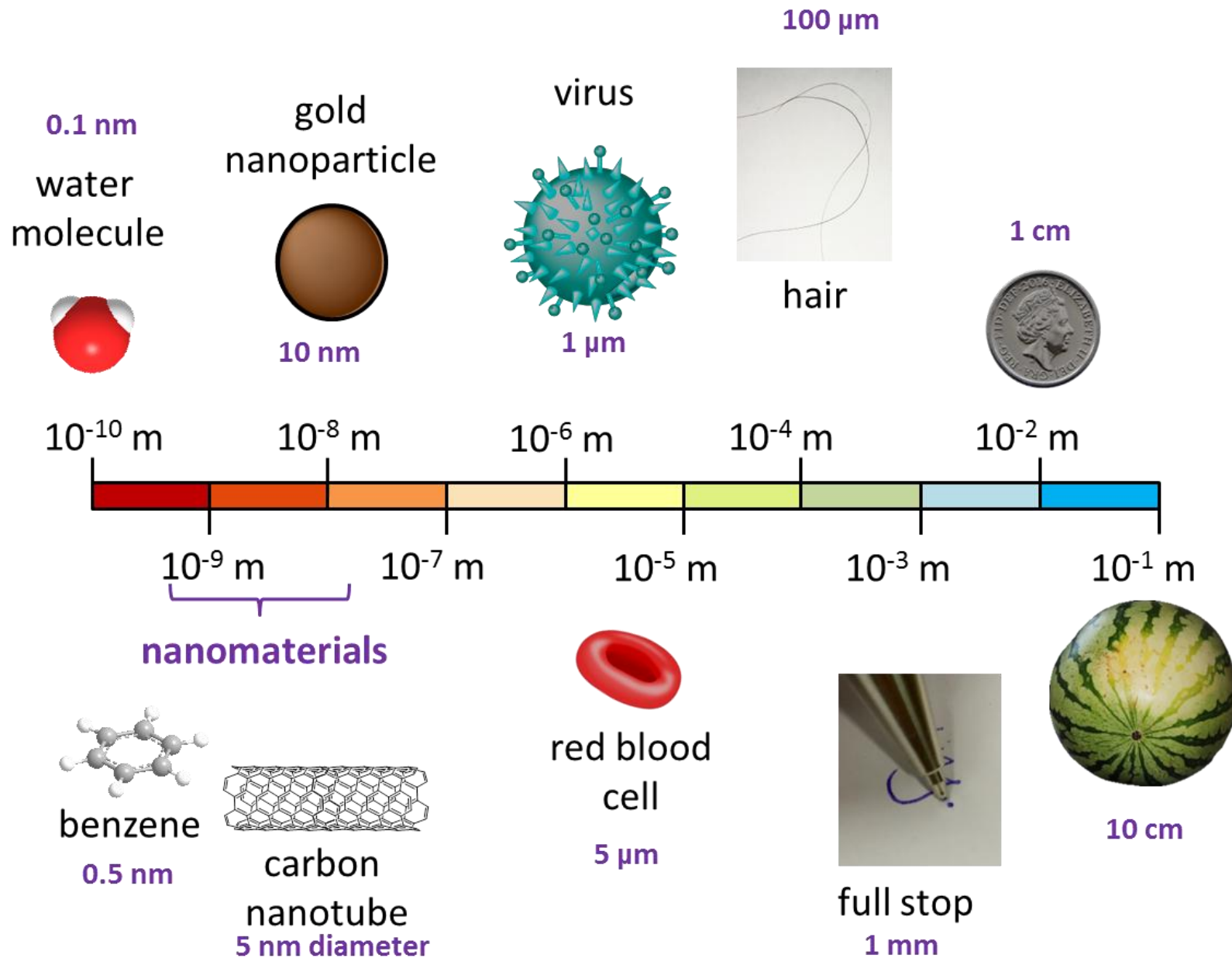
Introduction to Nanotechnology

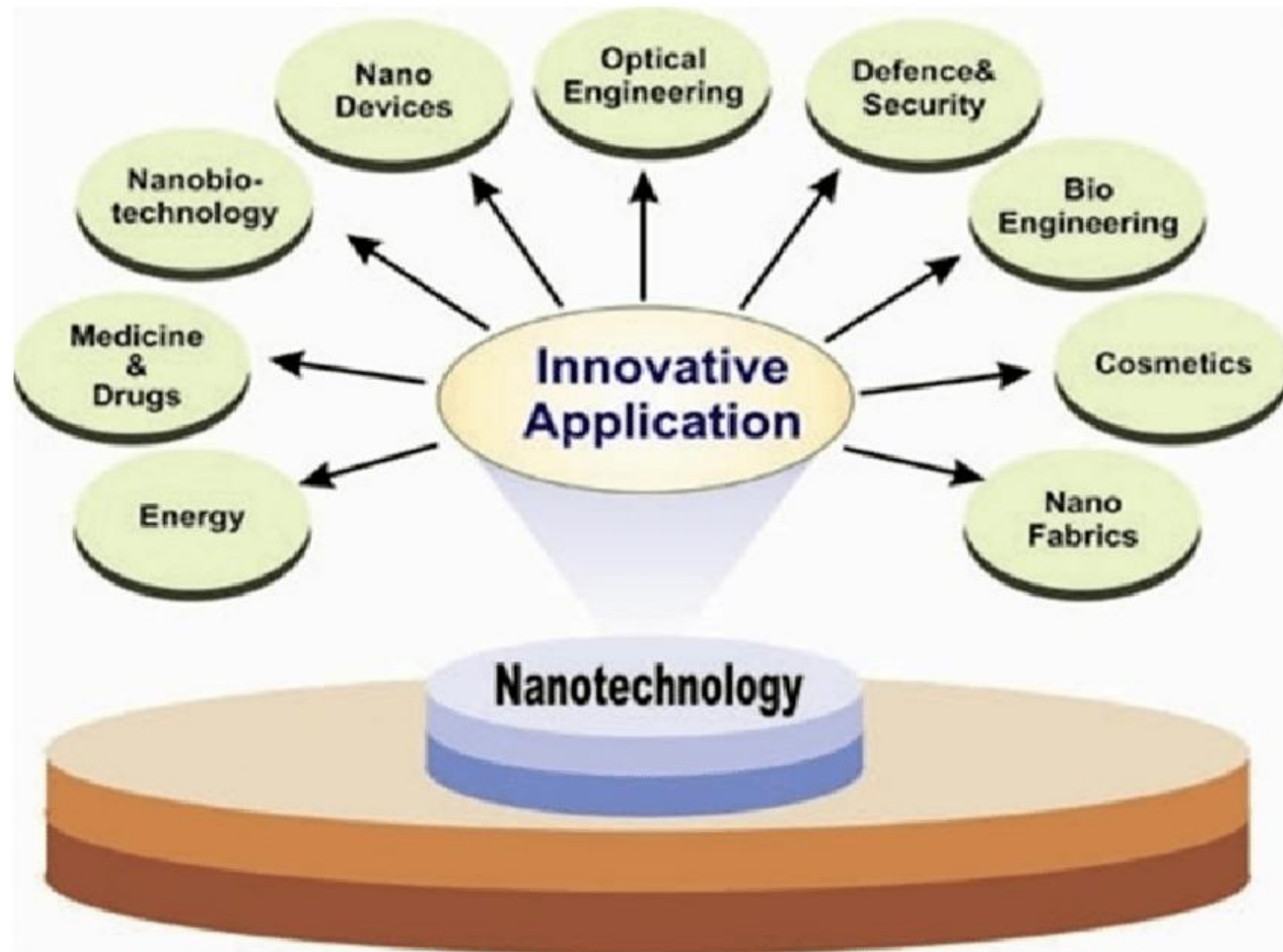
INTRODUCTION

The linguistic form **nano** originates from the classical Latin **nanus** or its ancient Greek etonym **nanos**, meaning “dwarf”

Nanomaterials are typically between **1 and 100 nanometres** (nm) in size – with 1 nm being equivalent to one billionth of a metre (10^{-9} m).

Comparison of nano size with other lengths





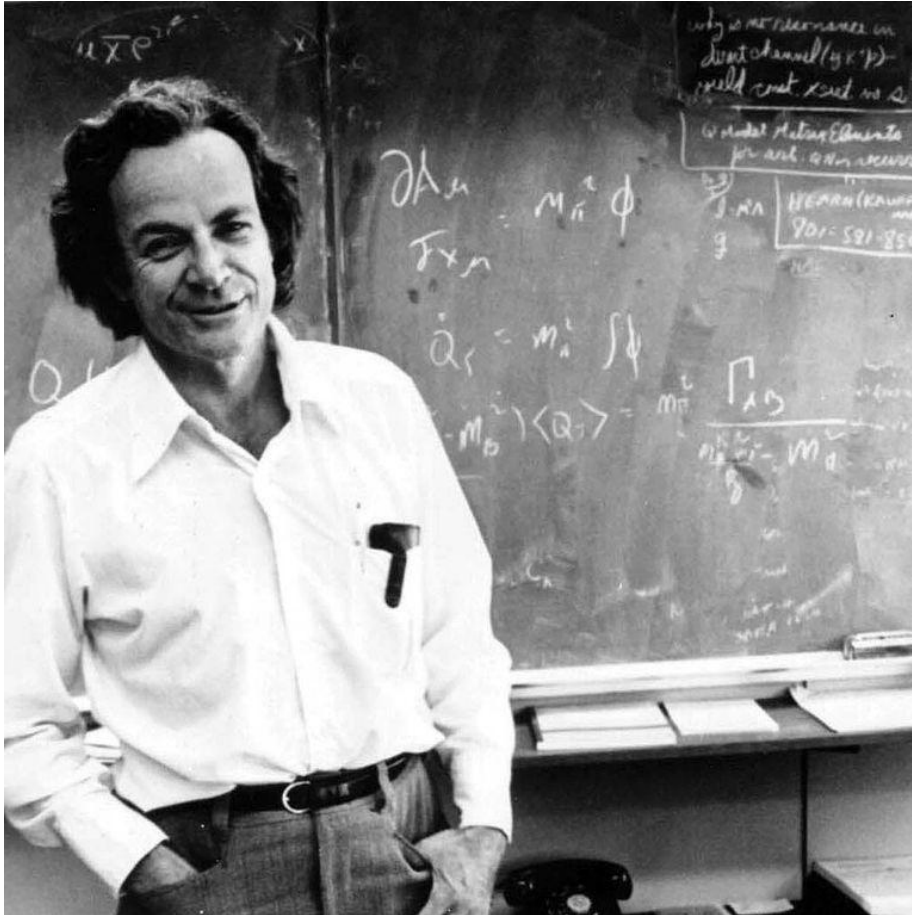
HISTROY

1857: **Michael Faraday discovered colloidal “ruby” gold**, demonstrating that nanostructured gold under certain lighting conditions produces different-colored solutions.



HISTROY

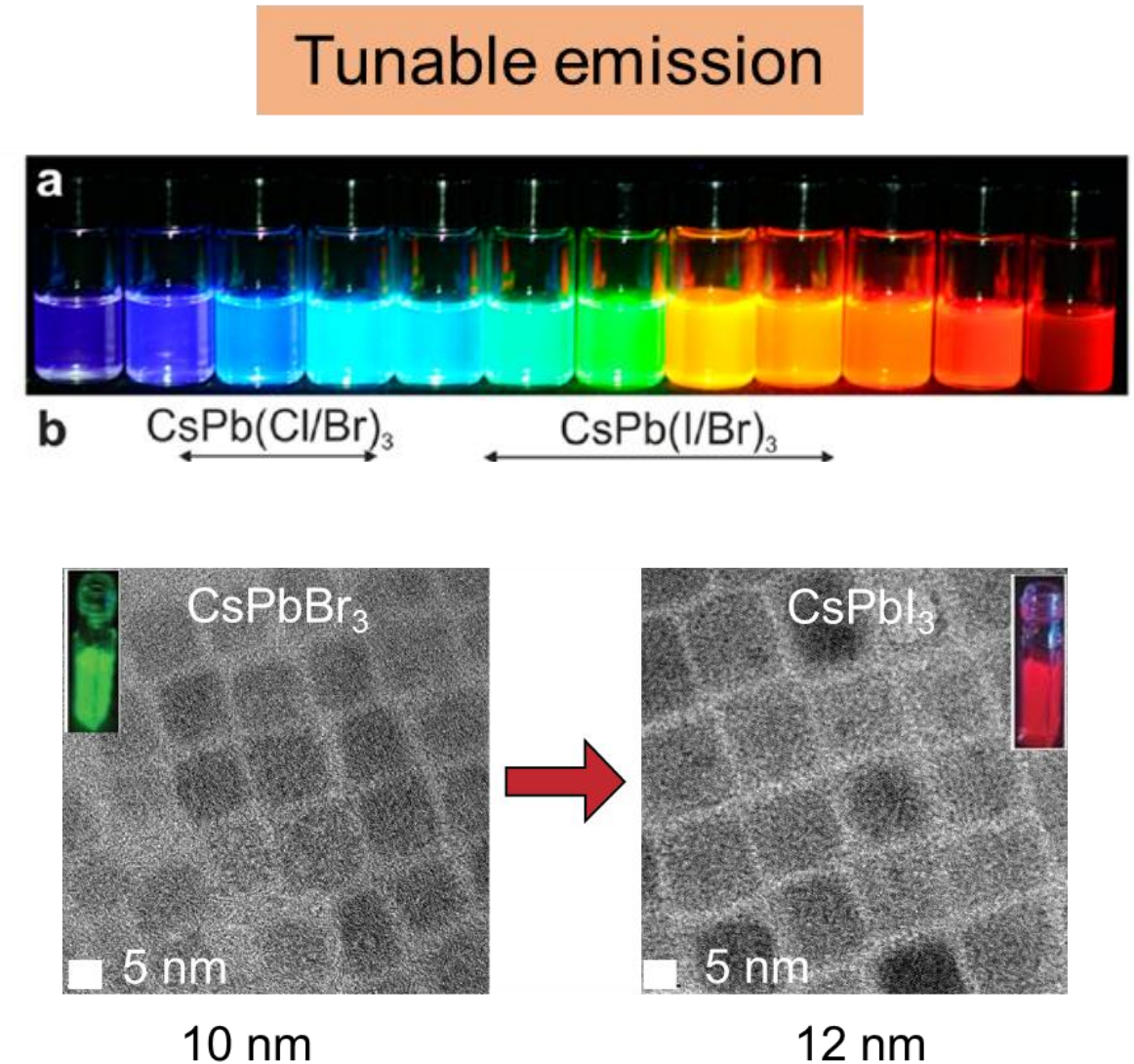
1959: Richard Feynman of the California Institute of Technology



"There's Plenty of Room
at the Bottom"

Size-Dependent Properties of Nanomaterials

- **Melting-Point Depression:** Gold nanoparticles (≈ 2.5 nm) melt at ~ 300 °C vs. 1064 °C bulk.
- **Quantum Confinement (Optical Properties):** CdSe quantum dots: size determines emission color (blue vs. red). Silicon quantum dots: photoluminescence range from yellow (~ 600 nm) to near-infrared (~ 1000 nm).
- **Plasmonic Effects:** Gold nanoparticles: color shifts (red to yellowish) as size drops; resonance shifts; quantum behavior dominates < 3 nm.

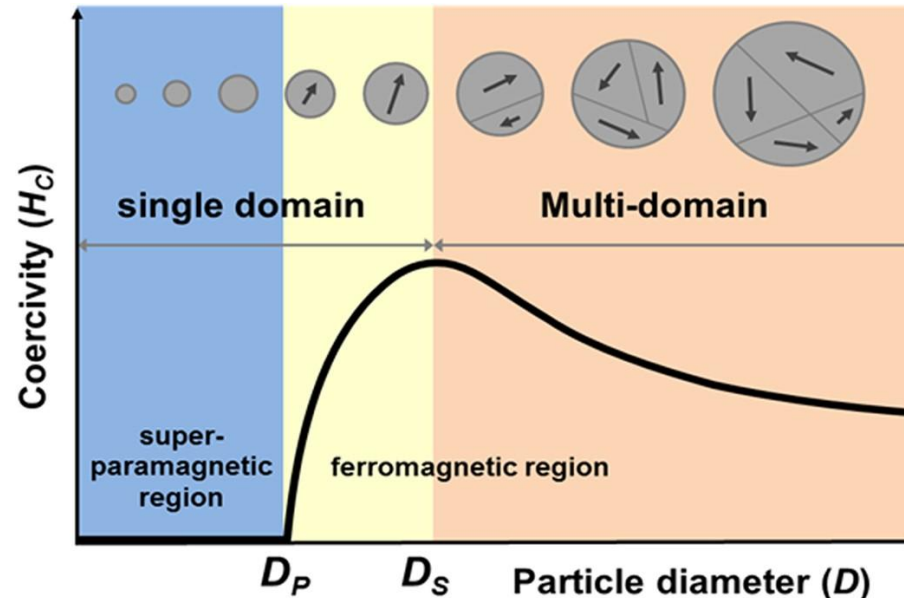


Size-Dependent Properties of Nanomaterials

Electrical properties:

- ❑ *Metal nanoparticles* can show tunable semiconducting properties depending on the particles size.
- ❑ *$Y_2Ir_2O_7$ Nanoparticles*: Compared to bulk, these nanoparticles exhibit a colossal increase in electrical conductivity due to enhanced double exchange interactions and shifts in oxidation state dynamics.
- ❑ *Graphene*: Display exceptionally high transport efficiency—electrons can travel micrometers without scattering.

Magnetic Behavior: Magnetic nanoparticles (<10 nm) display superparamagnetism, useful for switching applications.

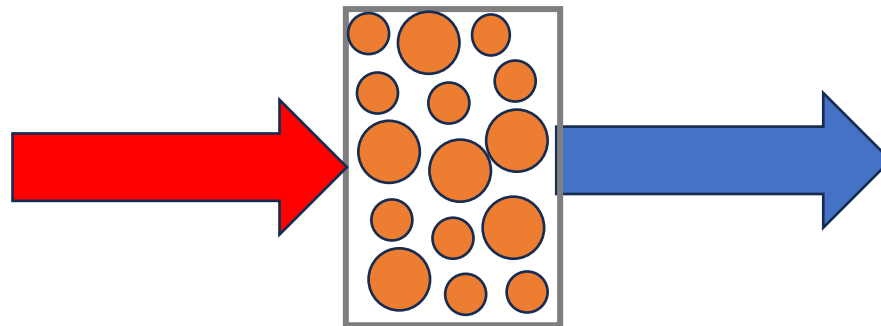


Size-Dependent Properties of Nanomaterials

Superconductivity: Indium (In) Nanowire, Lead (Pb) Nanowires, Single-Walled Carbon Nanotubes (SWCNTs), NiBi₃-Coated Ni Nanoparticles....shows superconducting nature.

Mechanical & Chemical Strength: Nanophase ceramics with grain size <100 nm show elevated hardness and toughness.

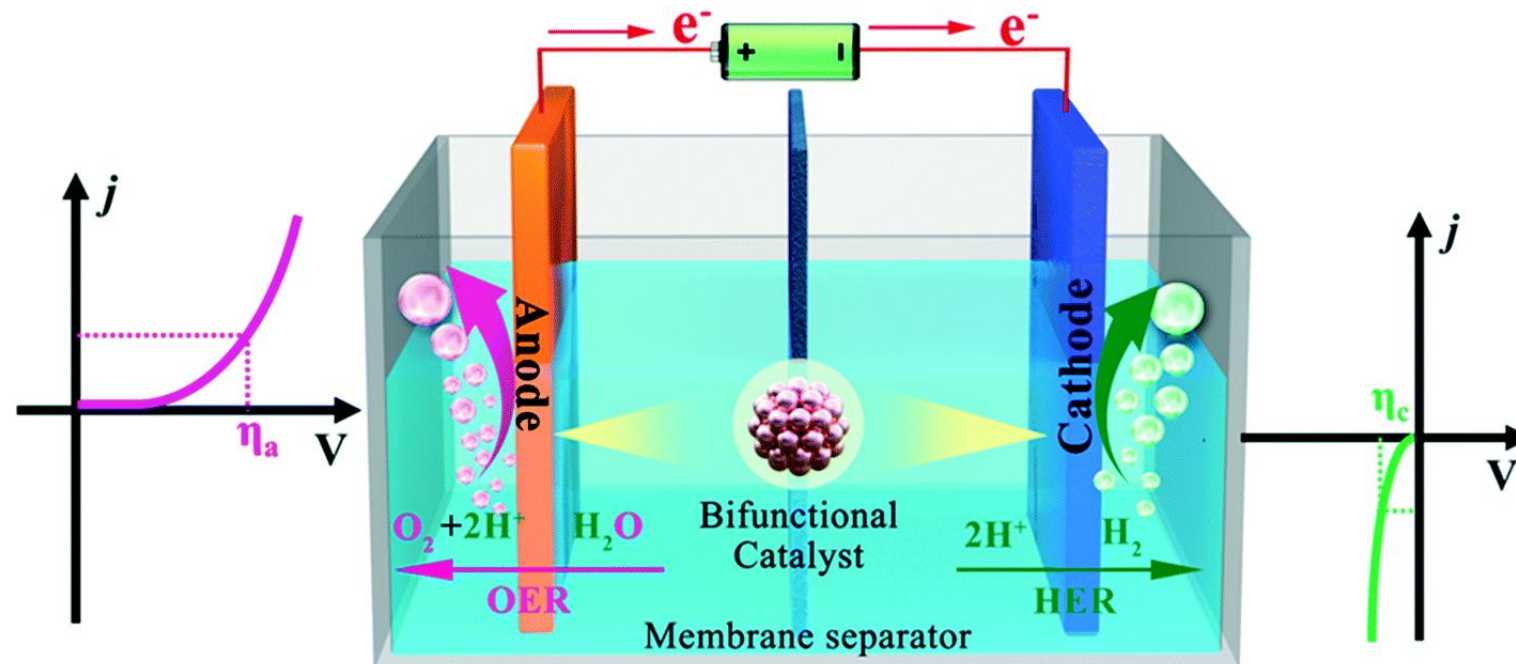
Nonlinear Optical Properties: Copper nanoparticles (5–100 nm) show highly size-sensitive hyperpolarizability (β).



Size-Dependent Properties of Nanomaterials

Chemical properties:

- ❑ Bulk gold is chemically inert, but when shrunk to nanoclusters, it becomes catalytically active due to increased surface sites and electronic alterations.
- ❑ Catalytic activity increases for most of the nanoparticles due to the higher surface to volume ratio.



Differentiation between bulk and nanoscale behaviours

Property	Bulk materials	Nanomaterials
Surface-to-Volume Ratio	The ratio of surface atoms to interior atoms is negligible.	A significant portion of atoms are on the surface, dramatically increasing the ratio and leading to unique and more intense surface-level interactions.
Governing Physics	Follows classical Newtonian physics.	The behavior of atoms and electrons is governed by quantum mechanics, leading to novel size-dependent effects
Optical Properties	Light interaction is independent of size. For example, bulk gold appears yellow.	Light interaction is size-dependent due to quantum effects like surface plasmon resonance. Gold nanoparticles appear red, blue, or green depending on their size.
Electrical Properties	Electrical conductivity is a stable, bulk property.	Conductivity is size-dependent and can be influenced by quantum confinement. For instance, carbon nanotubes can be either conductors or semiconductors depending on their structure.

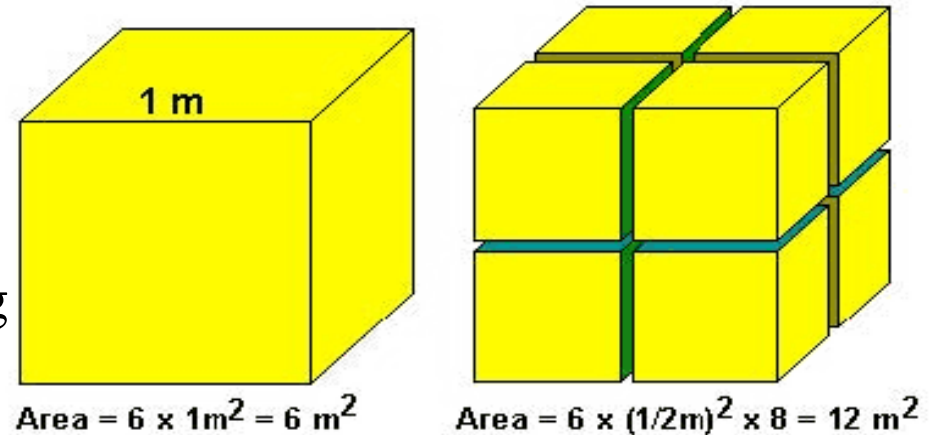
Differentiation between bulk and nanoscale behaviours

Property	Bulk materials	Nanomaterials
Melting Point	The melting point is a fixed material property.	Melting temperature decreases significantly as particle size is reduced. This is a result of surface atoms having lower cohesive energy.
Reactivity and Catalysis	Less reactive due to a small surface area for chemical reactions. Bulk gold is chemically inert	Highly reactive due to the large, highly active surface area. Gold nanoparticles are effective catalysts for various chemical reactions.
Magnetic Properties	Exhibit stable magnetic behavior (ferromagnetic, paramagnetic, etc.)	Can exhibit completely different magnetic behavior. Non-magnetic bulk materials like gold can become magnetic at the nanoscale.
Mechanical Properties	Strength is typically lower and often limited by grain boundaries and defects.	Can exhibit higher strength, toughness, and elasticity due to the scarcity of defects and high surface area.

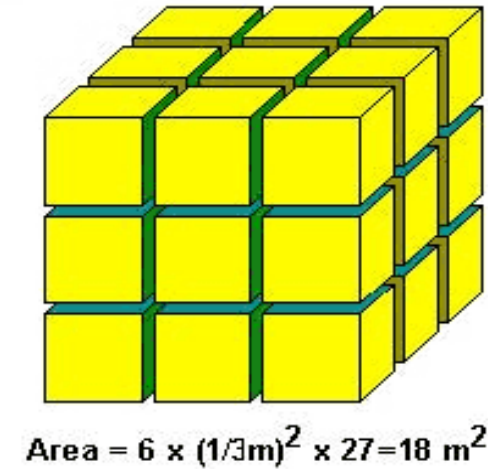
Surface to Volume Ratio Increases

As surface to volume ratio increases

- A greater amount of a substance comes in contact with surrounding material.



- This results in better catalysts, since a greater proportion of the material is exposed for potential reaction.



1) Sphere — find S/V at 50 nm radius

A spherical nanoparticle has radius $r = 50 \text{ nm} = 50 \times 10^{-9} \text{ m}$.

For a sphere: $\frac{S}{V} = \frac{3}{r}$.

$$\frac{S}{V} = \frac{3}{50 \times 10^{-9}} = 0.06 \times 10^9 = 6.0 \times 10^7 \text{ m}^{-1}.$$

2) Cube — find S/V at 30 nm edge

Cube with edge $a = 30 \text{ nm} = 30 \times 10^{-9} \text{ m}$. For a cube: $S/V = 6/a$.

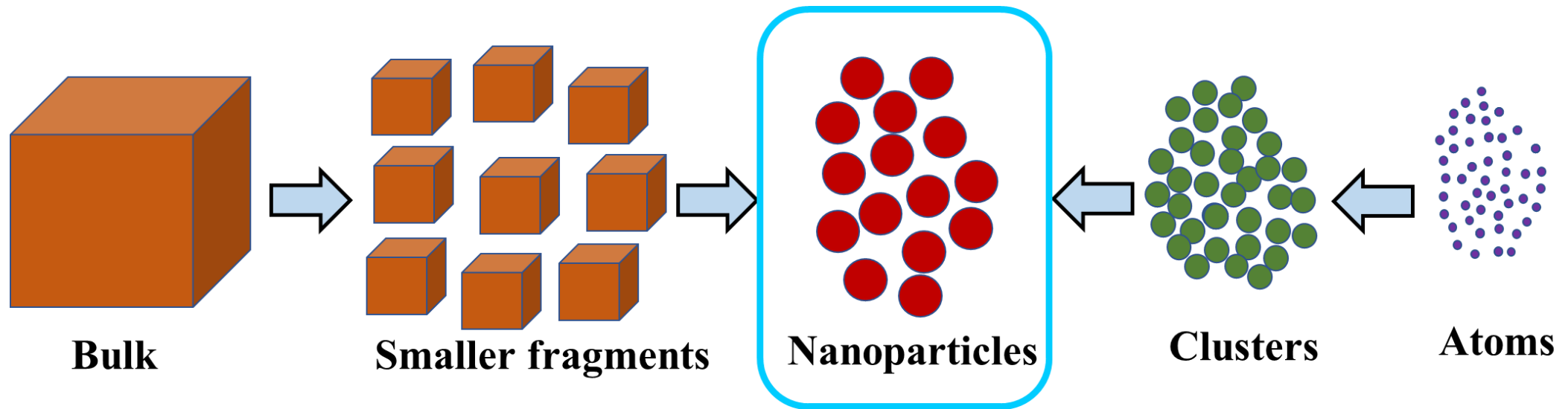
$$S/V = \frac{6}{30 \times 10^{-9}} = 0.2 \times 10^9 = 2.0 \times 10^8 \text{ m}^{-1}.$$

Answer: $2.0 \times 10^8 \text{ m}^{-1}$. [Wikipedia](#)

SYNTHESIS OF NANOMATERIALS

Top-Down

Bottom-Up



Top-Down Approach

Physical reduction of bulk material into nanostructures.

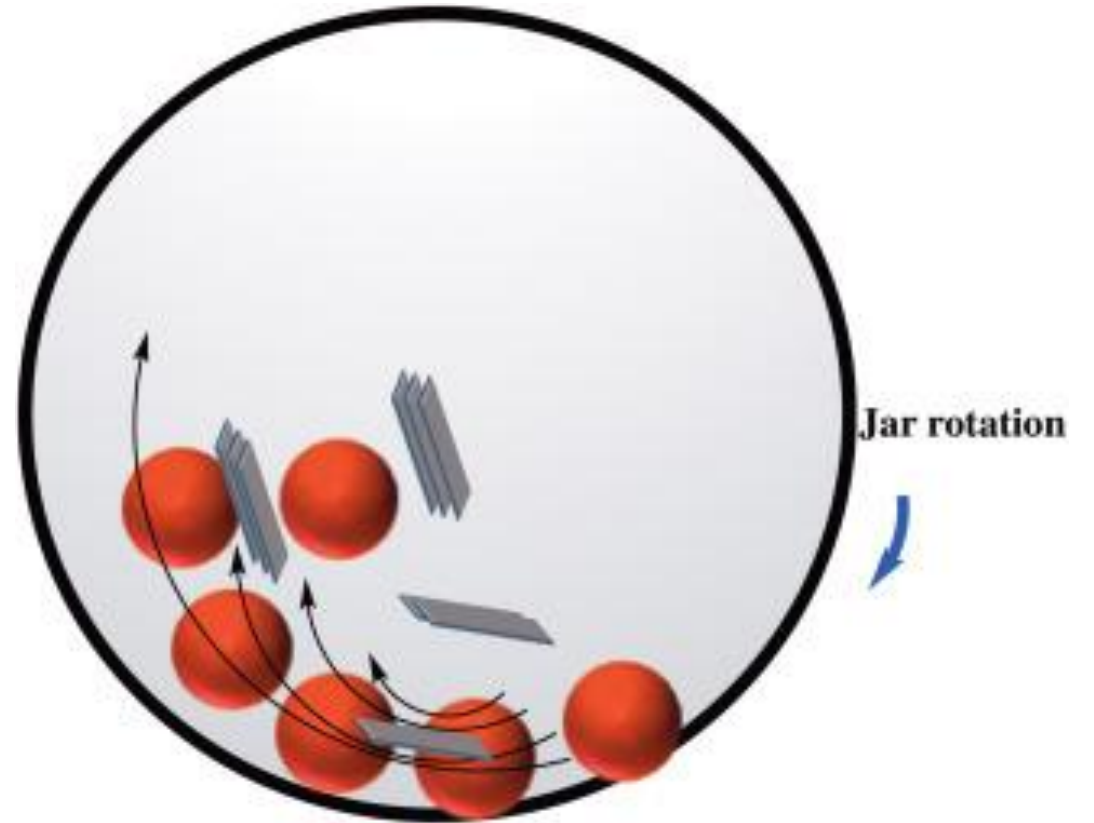
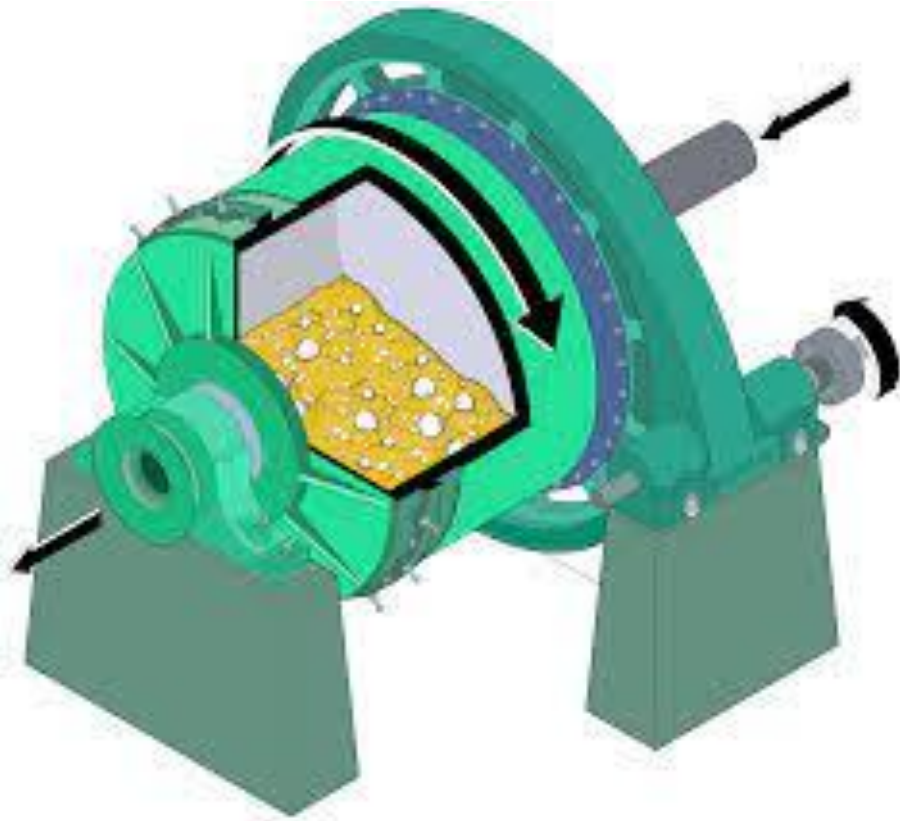
- **Methods:**

- Mechanical milling (ball milling)
- Lithography (photolithography, e-beam lithography)
- Laser ablation
- Etching techniques

- **Advantages:** Simple, industrially scalable

- **Limitations:** Surface defects, poor control over size and shape

Ball Milling



Lithography

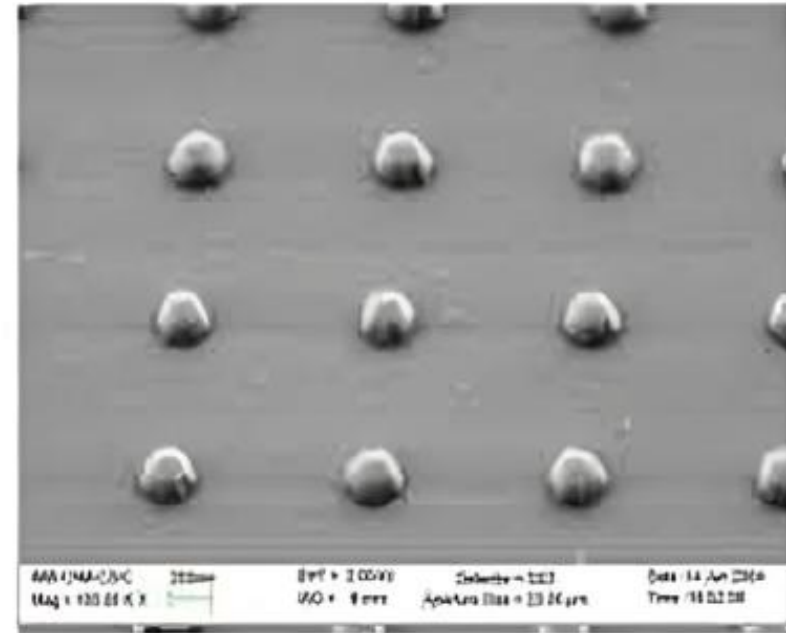
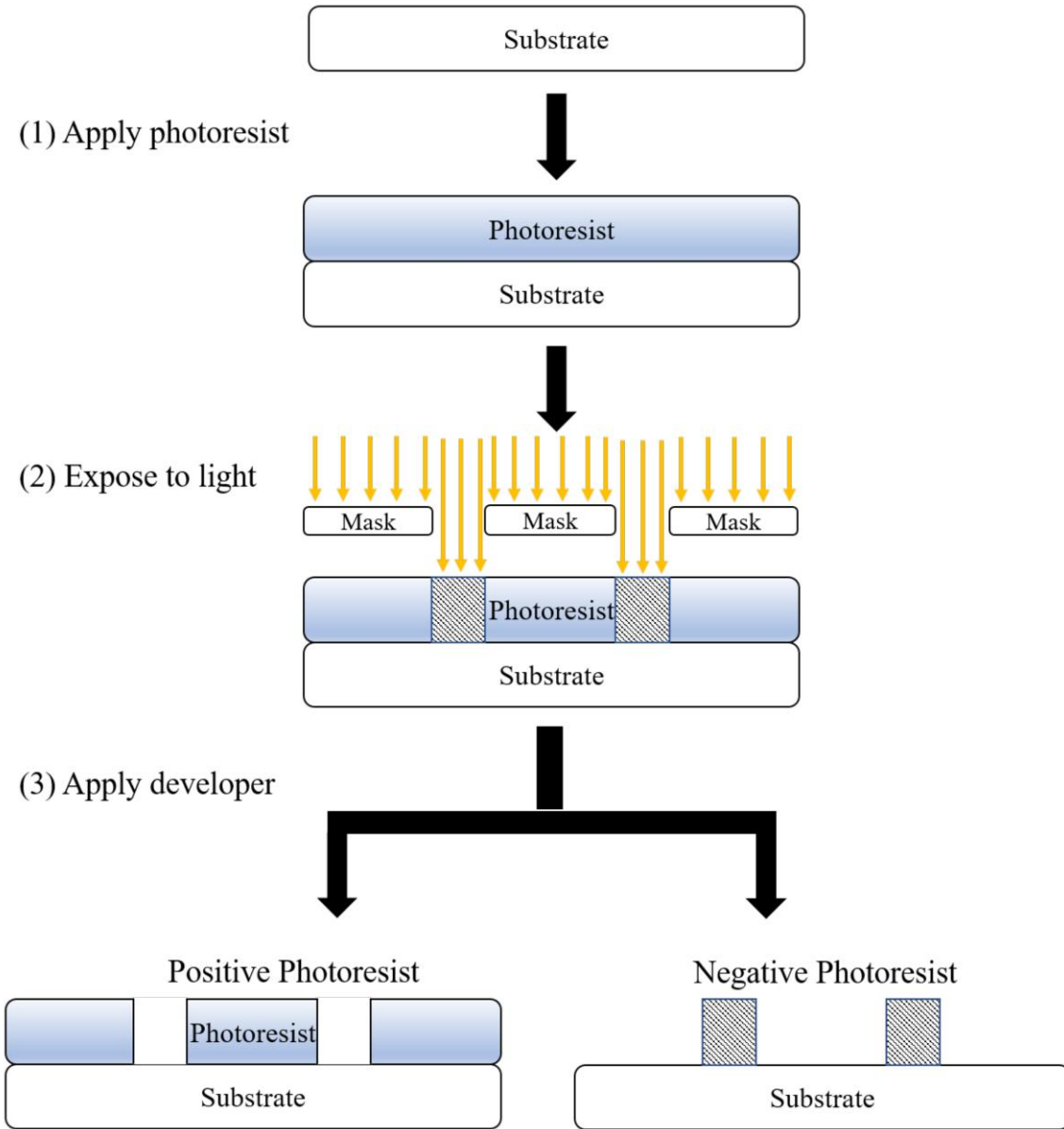
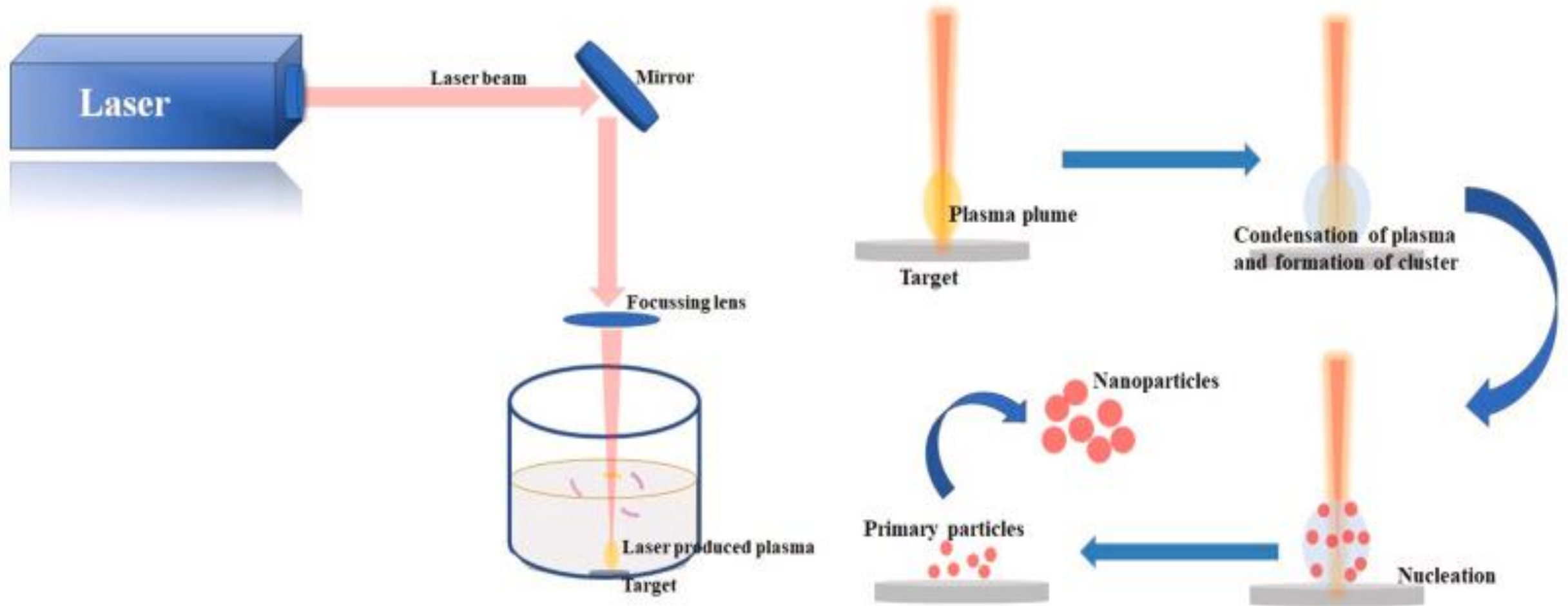
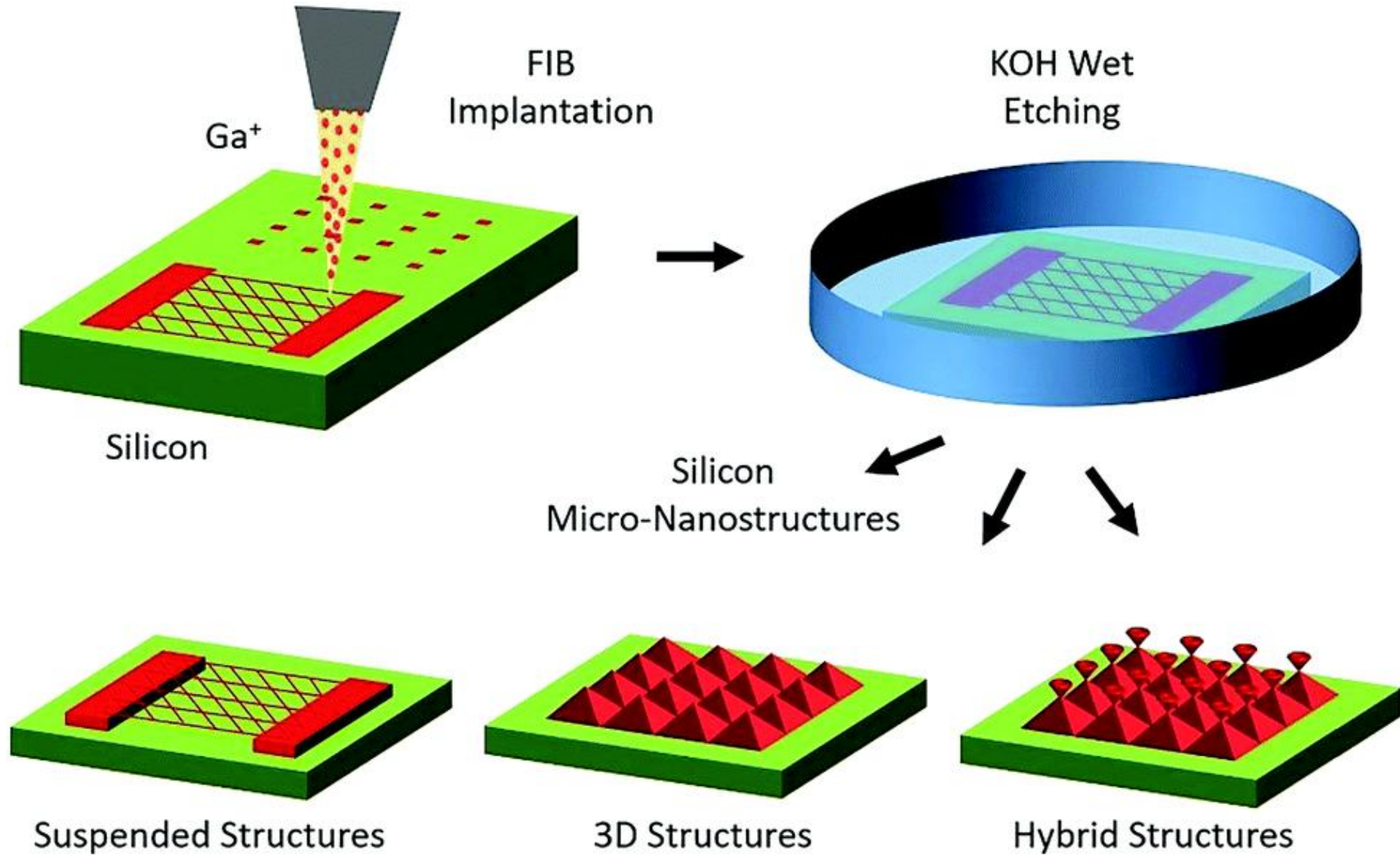


Figure: Photolithography process

Laser ablation



Etching techniques



Bottom-Up Approach

Atom-by-atom or molecule-by-molecule assembly.

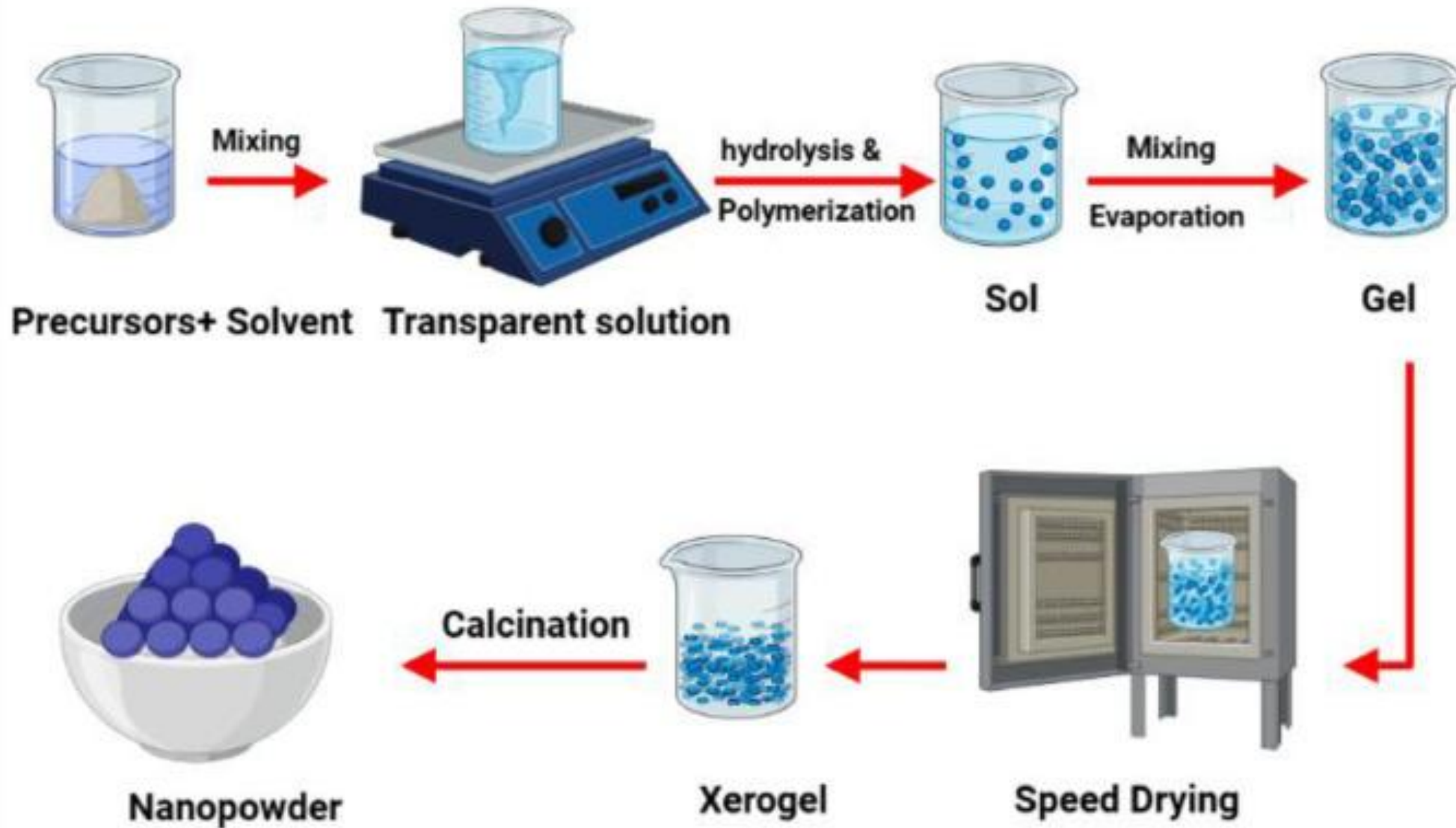
- **Methods:**

- Sol-gel synthesis
- Chemical vapor deposition (CVD)
- Atomic layer deposition (ALD)
- Hydrothermal / solvothermal synthesis
- Self-assembly

- **Advantages:** Better control of morphology, purity

- **Limitations:** Complex process control, costly chemicals

Sol-gel method



Hydrothermal/Solvothermal method

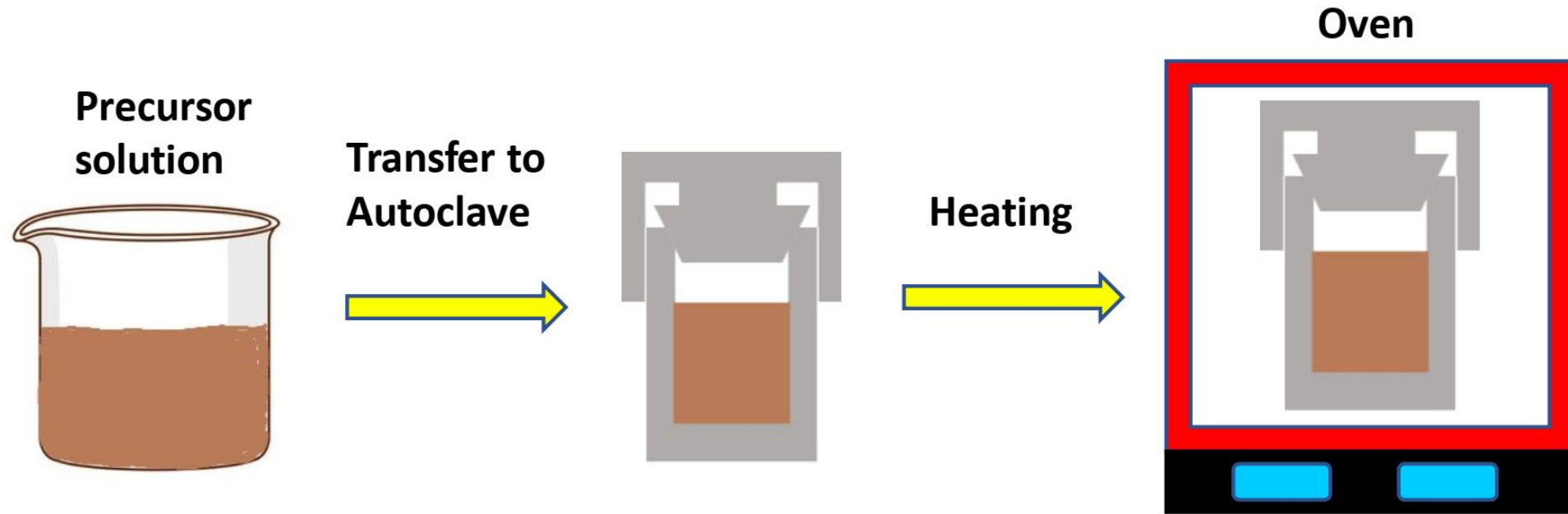
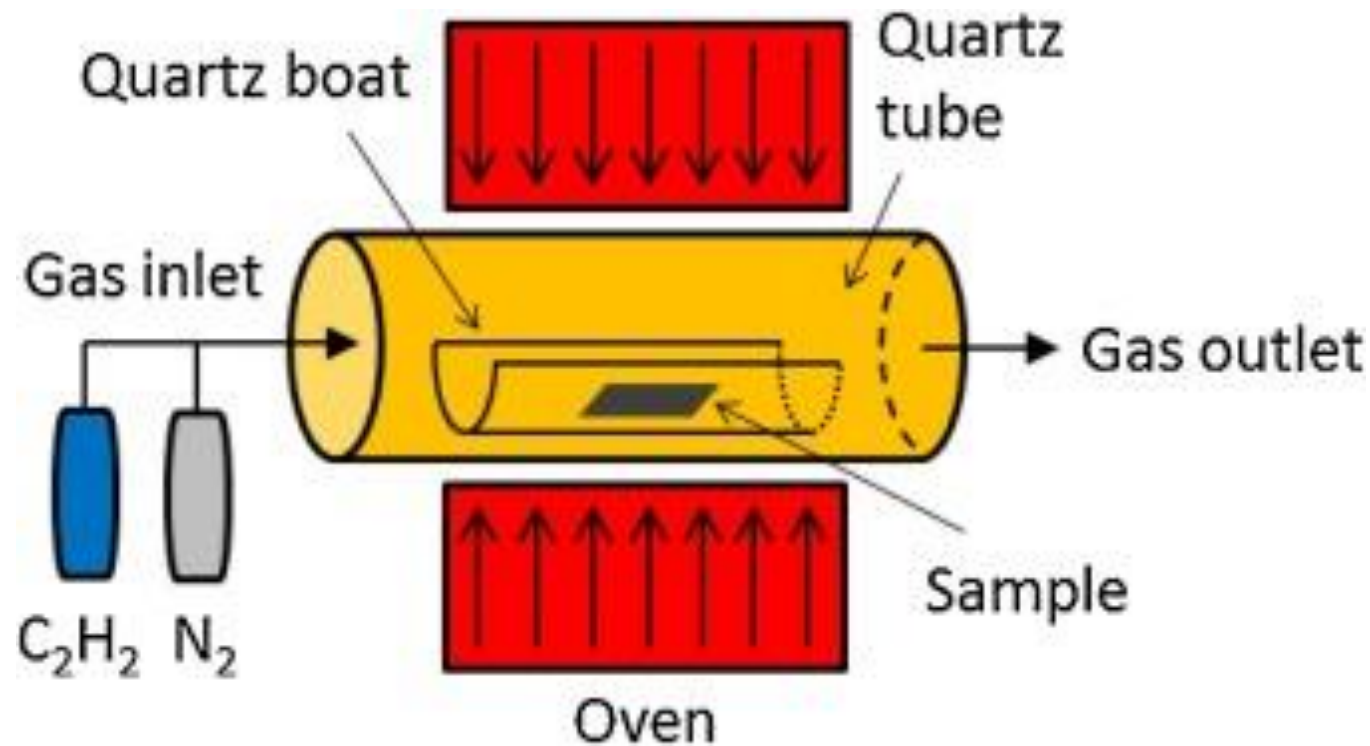


Figure 2.3. Schematic representation of nanoparticle synthesis process by hydrothermal process.

Chemical Vapor Deposition (CVD)

- Film forms by chemical reaction of gas precursors on substrate surface.
- Produces high-quality, uniform films.
- Suitable for semiconductor manufacturing.



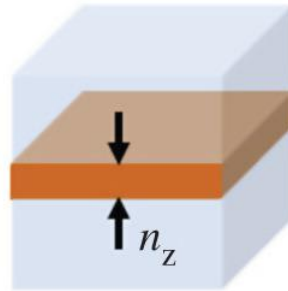
Quantum Effects at the Nanoscale

Quantum confinement

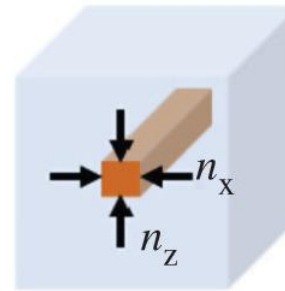
no confinement
bulk
(3D materials)



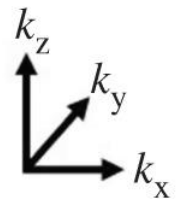
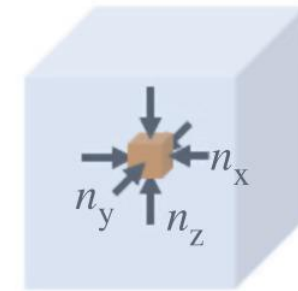
1D confinement
Q-well/ultrathin film
(2D materials)

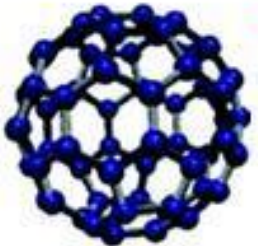
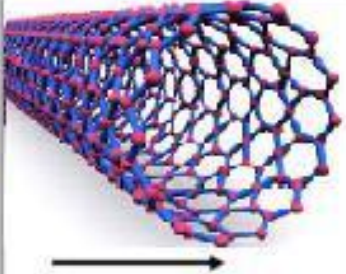
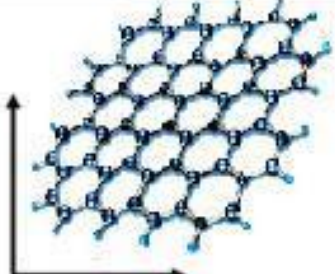
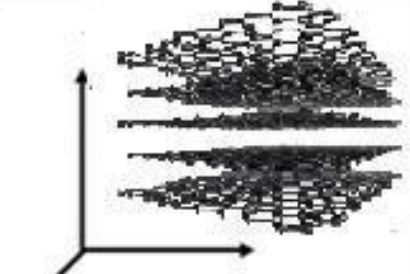


2D confinement
Q-wire
(1D materials)



3D confinement
Q-dot
(0D material)



0D	1D	2D	3D
			
Fullerene	Carbon Nanotube	Graphene	Graphite